



Khwaja Yunus Ali University
School of Science and Engineering
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering
OBE Course Outline

Part A

1. **Course Code:** CSE 1208
2. **Course Title:** Engineering Drawing and Auto CAD Lab
3. **Course Type:** Core
4. **Level:** 1.2
5. **Academic Session:** Spring-2015
6. **Course Instructor:** Sumaiya Shahria
7. **Prerequisites:** N/A
8. **Credits:** 1.5
9. **Credit hours/week:** 3
10. **Total Marks:** 100
11. **Course Rationale**

An engineering drawing is a subcategory of technical drawings. The purpose is to convey all the information necessary for manufacturing a product or a part. Engineering drawings use standardized language and symbols. This makes understanding the drawings simple with little to no personal interpretation possibilities. Computer Aided Drafting is a process of preparing a drawing of an object on the screen of a computer. There are various types of drawings in different fields of engineering and sciences. In the fields of Computer Science and Engineering, the drawings of machine components and the layouts of them are prepared. The use of CAD process provides enhanced graphics capabilities which allows any designer to conceptualize his ideas, modify the design very easily and to perform animation using colors, fonts and other aesthetic features.

12. **Course Objectives:**

The main objective of the course are to further the ability to communicate information by graphical means, using also CAD software packages. This will be achieved through the ability to visualize and understand spatial relationships, and the competence to select and use appropriate graphical methods for representing design concepts. The course also aims to combine the essential parts of the Engineering Drawing with computer-aided drafting (CAD).

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Course Learning Outcome (CLO)		
CLOs	After successful completing this course, students will be able to.	PLO Mapping
CLO1	Acquire basic knowledge of drawing applications found in engineering and technology in general	PLO1, PLO2
CLO2	Perform three-dimensional computer-aided drafting (CAD)	PLO, PLO3
CLO3	Develop skill of representing design concepts and solutions	PLO3, PLO12

Part B

Week	Topic	Teaching Strategy	Assessment Technique	Alignment with CLOs
01	Introducing to technical drawing Specific outcomes: - Course overview & Introduction - Introduction to Engineering drawing, Numbering and heading, Line types etc.	Power point, tutorial,	Presentation, Report, final exam, viva	CLO1
02	Dimensioning Practice Specific outcomes: - Plane geometry - Solid geometry - Developments and concept of true shapes and sections, - Concept of Projections	Power point, Hands on technology, Jigsaw technology	Lab experiment, Report, final exam, viva	CLO1
03	Geometric Construction and Projection of Points Specific outcomes: Oblique Drawing of objects	Power point, tutorial, Case studies	Report, final exam, viva	CLO1
04	Introducing to orthographic Drawing Specific outcomes: - Plan, elevation and section of engineering structures - Orthographic projections of objects	Power point, tutorial, Group discussion, Providing feedback	Report, Lab experiment, final exam, viva	CLO1
05	Introduction to Computer Aided Design (CAD) Specific outcomes: - How to start using AutoCAD - Creating folder Saving a file - Drawing simple Isometric Box	Power point, informal group, Modeling	Report, performance, final exam, viva	CLO2
06	Drawing object in different Plane Specific outcomes: - Top, Front and Right side plane drawing - Use of reference plane - Sketch with different tool of SOLIDWORKS software	Power point, tutorial, Brainstorming	Report, Concept map, final exam, viva	CLO1 CLO2
07	3D drawing with particular software Specific outcomes: - ORTHO, OSNAP, GRID, DYN, dimension, Plot & Create PDF, Polygon, Move, copy, Offset, Arc - Text style & Text - Use of different feature of AUTO CAD software.	Self-assessment, Think pair share	Report, final exam, Decision making task, viva	CLO2
08	Specific outcomes: - Definition and application of Gearbox - Use of different pattern - Advanced tool of Solid Works software	Groups evaluation	Report, final exam, viva	CLO2
09	Designing Nut Bolt Specific outcomes:	Power point, tutorial,	Presentation, Report, final exam, viva	CLO2

	• Circular pattern • Thread cutting • Helix angle selection			
10	Designing assemble with 3D CAD software. Specific outcomes: • Mate surface • Assembling • Creating animation	Power point, Hands on technology, Jigsaw technology	Lab performance, Report, final exam, viva	CLO3
11	The production of assembly drawings from detail drawings. Specific outcomes: • Components of engines, • compressors, pumps etc.	Power point, tutorial,	Presentation, Report, final exam, viva	CLO2
12	Layout and Presentation Specific outcomes: • Sheet layout of drawings to modern drawing office practice, • Including blocks for items such as title, date, drawing and job number	Power point, Hands on technology, Jigsaw technology	Lab experiment, Report, final exam, viva	CLO3
13	The selection and preparation of drawing paper Specific outcomes: • subsequent reproduction	Power point, tutorial, Case studies	Report, final exam, viva	CLO2 CLO3
14	Drafting Specific outcomes: • Preparing blueprint from the drawn object	Power point, tutorial, Group discussion, Providing feedback	Report, Lab performance, final exam, viva	CLO3
15	Designing Project Specific Outcome: • 3 to 4 Group project drawing based on relevant area	Power point, informal group, Modeling	Report, performance, final exam, viva	CLO3

14.

Part C

15. Assessment and Evaluation

1. Assessment Strategy

Students are evaluated out of total 100 Marks. There are different types of assessment tools. In the **Final Examination**, total marks is **60** and allotted time is **2 Hours**. The detail with **CLOs-Assessment Mapping** is given below.

	CIE-Continuous Internal Evaluation (40)			SEE-Semester End Examination (60)	
Bloom's level	Assignment(10)	Lab performance (10)	Report (20)	Final Exam (50)	Viva (10)
Remember					2
Understand	5	2		10	2
Apply		2	5	10	2
Analyze		2	5	10	2
Evaluate		2	5	10	2
Create	5	2	5	10	

2. Make up procedures:

For weak students, we take extra class before their midterm and final examinations. Beside this, we counsel them and solve their problem as soon as possible.

Part D

16. 1) Recommended Reading

- Lab Reports
- Lab Manual

2) Supplementary Reading

- Text book of Engineering Drawing by K Venkata Reddy

3) Others

- Tutorial from different online source.